

Dibyakanti Kumar

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Github: Dibyakanti

Education

Jul 2024 – **The University of Manchester, UK**

Present *Ph.D in Computer Science*

Advisor : Prof. Anirbit Mukherjee, Prof. Alex Frangi

Jul 2018 – **Indian Institute of Technology, Guwahati**

Jul 2022 *Bachelor of Technology in Electronics and Electrical Engineering*

Minor in Computer Science

CGPA – 8.31/10

CGPA – 8.80/10

Experience

Dec 2022 – **Research Intern** UNIVERSITY OF MANCHESTER

Jul 2024 *Mentored by Prof. Anirbit Mukherjee*

- Assessed the efficacy of neural networks in solving **Partial Differential Equations (PDEs)** exhibiting finite-time blowup.
- Conducted rigorous **theoretical analysis** of existing neural PDE-solving frameworks to identify failure modes and underlying limitations.

Aug 2022 – **Software Developer** BARCLAYS

Jun 2024 **Market Risk** | C++ Developer | Apr'23 – Jun'24

- Maintained and extended the core framework used to compute Value-at-Risk (VaR) across multiple market indices.
- Improved **cache efficiency** by eliminating redundant bulk requests and implementing multi-threaded processing.
- Boosted Solace queue throughput by **reducing message redundancy**, lowering latency in the risk pipeline.
- Logging and Monitoring** | Python Developer | Aug'22 – Mar'23
 - Redesigned the logging and monitoring architecture to support all log types across the platform.
 - Built an **ML-based anomaly detection** system for log analysis, triggering proactive alerts to reduce the risk of major service interruptions.

Jun 2020 – **Research Intern** UNIVERSITY OF UTAH

Aug 2022 *Mentored by Dr. Vivek Gupta*

- Extended the semi-structured inference dataset **InfoTabS** through a semi-automatic, rule-based pipeline.
- Designed **intra-domain counterfactual tables** to discourage **BERT-class models** from exploiting spurious correlations and relying on memorized pre-training knowledge.
- Enforced **domain-specific constraints** to ensure the validity and consistency of generated tables.
- Improved downstream performance on **InfoTabS** by using the extended dataset for data augmentation.

Jun 2021 – **Software Developer Intern** BARCLAYS

- Jul 2021
- Modernized a legacy data ingestion framework by auditing and replacing obsolete libraries with actively maintained alternatives.
 - Parallelized data-processing pipelines in Python using **Dask** and **multiprocessing**, eliminating key performance bottlenecks.
 - Boosted data ingestion throughput by **6x**, reducing processing time and improving pipeline reliability.

Publications

- Generalization Bounds for Physics-Informed Neural Networks for the Incompressible Navier-Stokes Equations
S. Andre-Sloan, **D. Kumar**, A. F Frangi and A. Mukherjee [Paper]
- Convergent Stochastic Training of Attention and Understanding LoRA
Z. Sun, **D. Kumar**, A. F Frangi, A. Mukherjee and M. Sun [Paper]

- [3] Langevin Monte-Carlo Provably Learns Depth Two Neural Nets at Any Size and Data
D. Kumar, S. Jha and A. Mukherjee
 Accepted at **TMLR 2026** [Paper]
- [4] Towards Size-Independent Generalization Bounds for Deep Operator Nets
 P. Gopalani, S. Karmakar, **D. Kumar** and A. Mukherjee
 Published at **TMLR 2024** [Paper]
- [5] Investigating the Ability of PINNs To Solve Burgers' PDE Near Finite-Time BlowUp
D. Kumar and A. Mukherjee
 Published at **IOP-MLST journal** and short version at **NeurIPS 2023** ML4PS workshop [Paper]
- [6] Realistic Data Augmentation Framework for Enhancing Tabular Reasoning. **EMNLP 2022** in Findings.
D. Kumar, V. Gupta, S. Sharma and S. Zhang
 Findings of **EMNLP 2022** [Paper] [Website]

Skills

Languages Python, C++, Julia
 Frameworks JAX, PyTorch
 Utilities Docker, Git

Relevant Courses

Mathematics Linear Algebra, Multi-variable Calculus, Probability and Random processes, Graphs and Matrices
 ML Pattern Recognition and ML, Natural Language Processing, Data-Driven System Theory
 CS & Others Game Theory, Discrete Mathematics, Data Structure and Algorithm, Operating System, Information Theory, Network Coding and Application, Error Correcting Codes

Services

Organizer **DRSciML 2025 Workshop**
 Co-organized the DRSciML 2025 workshop
 Reviewer AISTATS, ICML, ICLR, Neurocomputing Journal, IOP-MLST Journal, TMLR
 Opensource **IITG.AI 2020-21**
 Head AI and ML community at IIT Guwahati

Achievements

2020 Recipient of **Silver Medal** at Inter-IIT TechMeet DataScience Competition
 2018 Ranked among the **top 2%** in JEE Advanced held for 0.15 million candidates